Calcurion Maths Worksheets

GCSE Mathematics

Topic: Quadratic Equations

MARKSCHEME

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1. $x = 2, 3$ **M1A1** [3]
2. $x = -3, -4$ **M1A1** [3]
3. $x = \pm 3$ **M1A1** [2]
4. $x = 0, 4$ **M1A1** [2]
5. $x = 3, -\frac{1}{2}$ **M1A1** [4]
6. $x = \frac{2}{3}, -1$ **M1A1** [4]
7. $(x+3)(x+5)$ **M1A1** [2]
8. $(x-4)(x-5)$ **M1A1** [2]
9. $(x-4)(x+4)$ **M1A1** [2]
10. $(2x+1)(x+3)$ **M1A1** [3]
11. $x = 2 \pm \sqrt{5}$ **M1A1** [3]
12. $x = \dfrac{-3 \pm \sqrt{17}}{4}$ **M1A1** [4]
13. $(x+3)^2 - 4$ **M1A1** [3]
14. $(x-4)^2 - 13$ **M1A1** [3]
15. $\Delta = 1$; two distinct real roots **M1A1** [3]
16. $x = 3 \pm \sqrt{5}$ **M1A1** [4]
17. Min $= 3$ at $x = 2$ **M1A1** [3]
18. Max $= 6$ at $x = -1$ **M1A1** [4]
19. Intercepts $(-3, 0), (2, 0), (0, -6)$ **M1A1** [4]
20. Intercepts $(-2, 0), (4, 0), (0, -8)$; TP $(1, -9)$ **M1A1** [5]
21. $k = 1$ **M1A1** [3]
22. $k^2 \geq 16 \Rightarrow |k| \geq 4$ **M1A1** [4]
23. $x = 1, 3$ **M1A1** [5]
24. $x = \pm 1, \pm 2$ **M1A1** [5]
25. $(x+2)^2 + 1 \geq 1 > 0$ **M1A1** [4]